Statistical Modeling to Predict Student's Dropout and Academic Success

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Project Motivation and Research Questions

WHY THIS TOPIC?

Student dropout and academic failure have high costs for students and institutions:

- lost tuition revenue
- delayed career progress
- lower graduation and retention rates

Using data-inclined approaches, Universities can:

- Boost Graduation Rates
- Optimize Resource Allocations
- Develop Evidence-Based Policies

KEY STUDY QUESTIONS

1. What are the biggest contributors to student dropout and academic failure?
2. Can we predict whether a student will drop out, remain enrolled, or graduate?
3. Can we predict dropout using only first-semester data?
4. How does enrollment age affect graduation likelihood?

Comprehensive Dataset Overview

4,424 Student Records · Polytechnic Institute of Portalegre · 2008–2019

DEMOGRAPHICS

- Age at enrollment
- Gender Distribution
- Marital Status
- National Diversity

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ACADEMIC PERFORMANCE

- First Year metrics
- Course Completion Rates
- Grade Averages
- Evaluation Outcomes

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SOCIOECONOMIC STATUS

- Family Occupation
- Family Education Levels
- Financial Aid Status
- Scholarship Recipients
- Debt indicators

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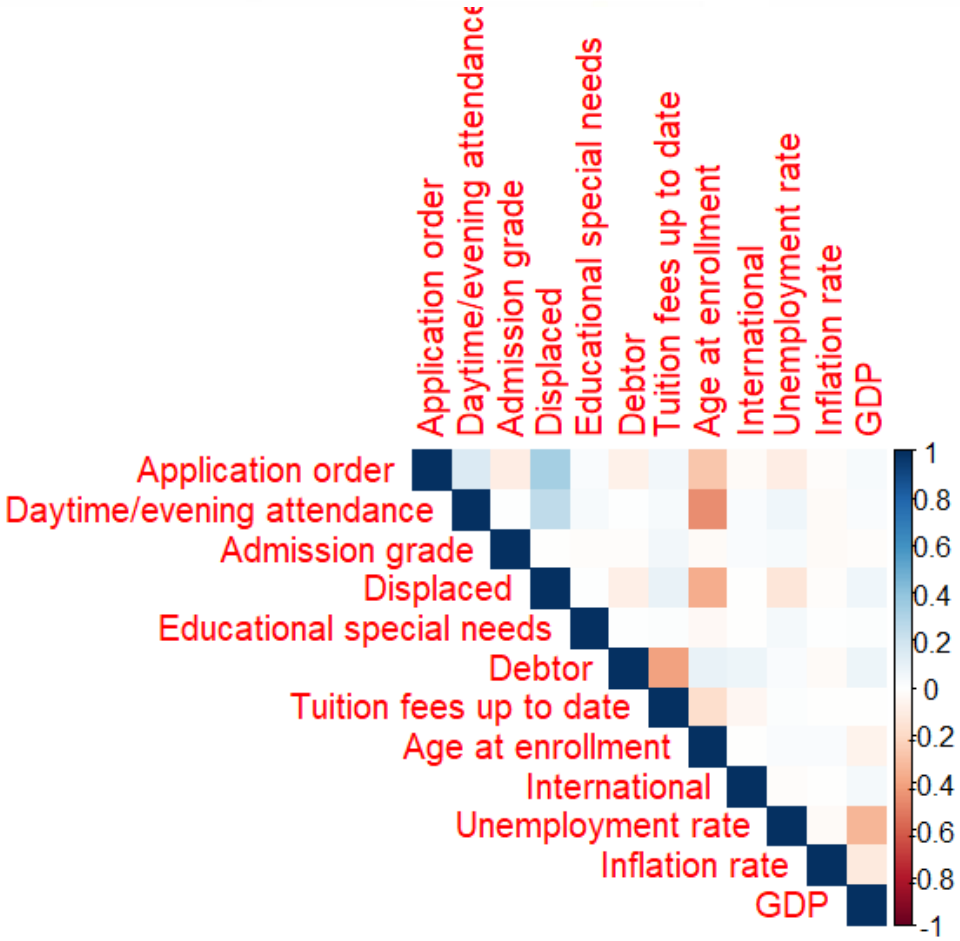
MACROECONOMICS

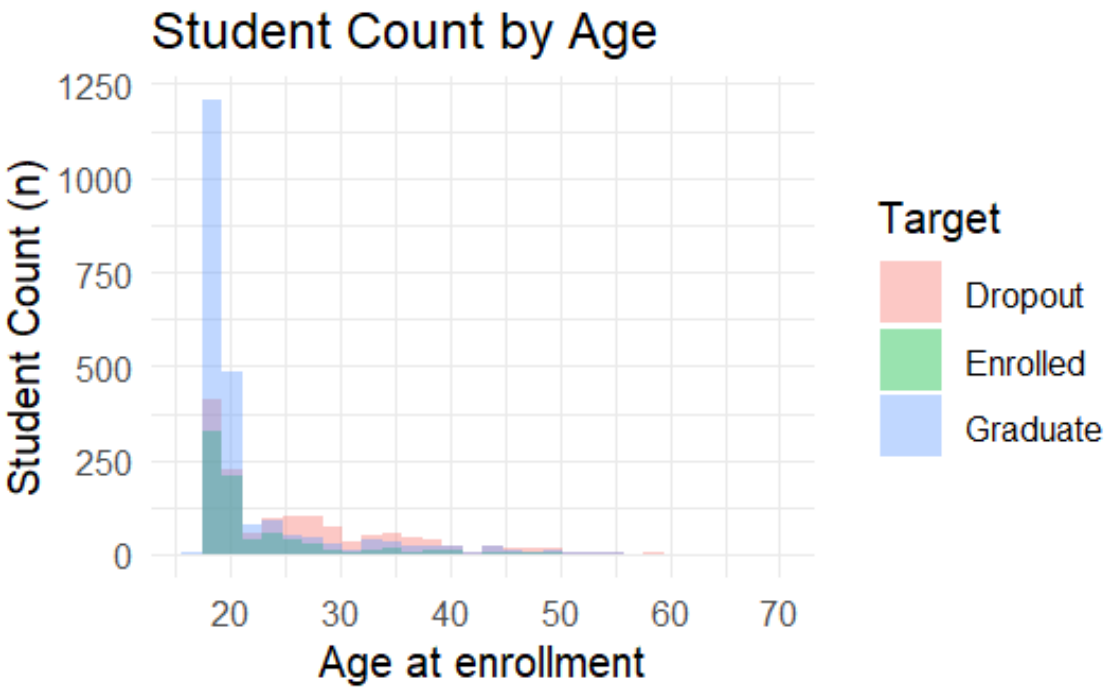
- National Unemployment Rates
- Inflation
- GDP

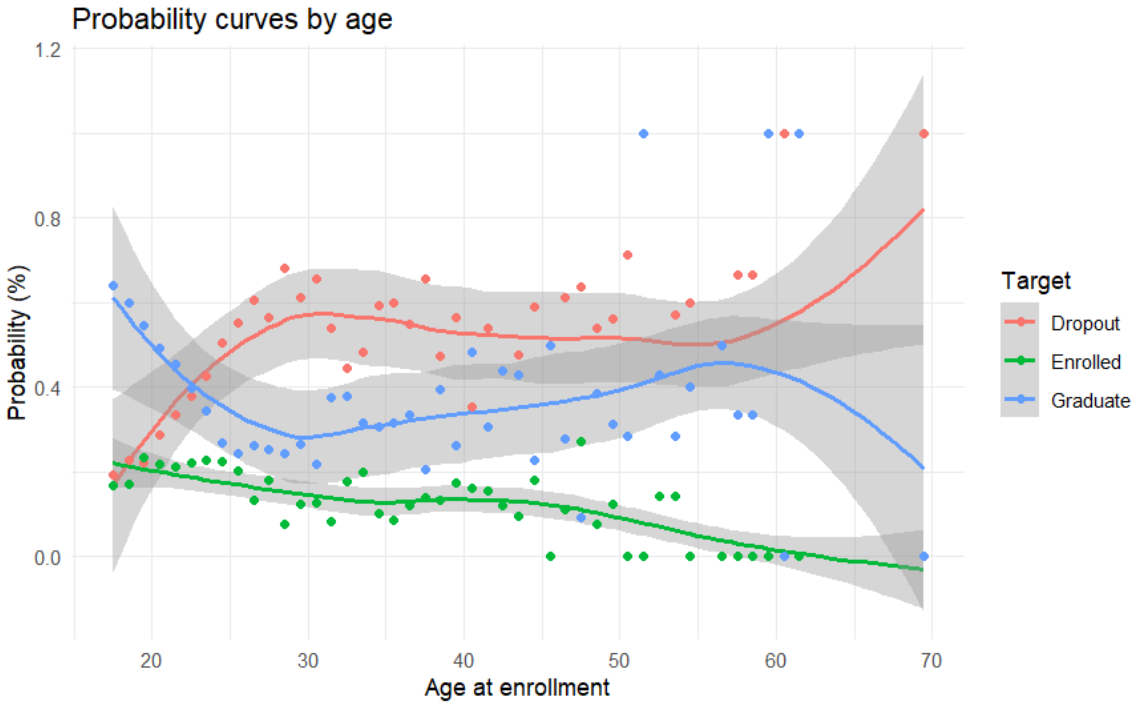
Data Source: UCI Machine Learning Repository

Variable Inspection – Correlation Matrix



Variable Inspection – Nonlinearity in Age





Methods: Multi Linear Regression

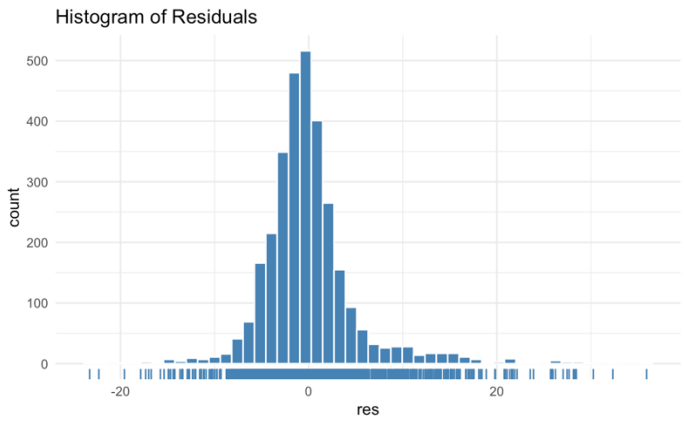
NOT A GOOD MODEL OF CHOICE

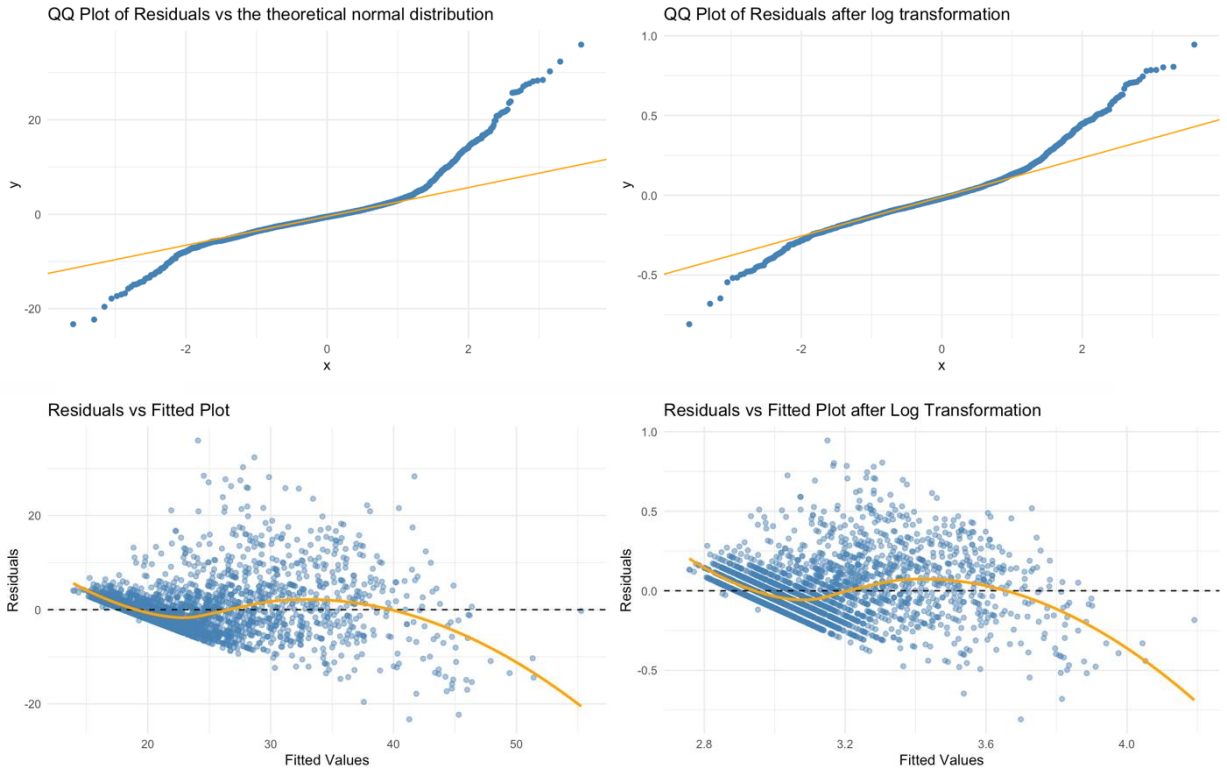
- Linear Regression -> BAD MODEL ($AdjR^2 = 0.57$)
  Model Split: 70/30

  Normality → ✗

  Constant Variance Assumption → ✗

  Why is not good fit → ✗





Current State of Knowledge

- Yakubu et al. (2025) – “Applying Machine Learning Approach to Predict Students’ Performance in Higher Educational Institutions”.
- Fahd et al. (2022) – “Application of Machine Learning in Higher Education to Assess Student Academic Performance, at-risk, and Attrition”.
- Tan & Shao (2015) – “Prediction of Student Dropout using Random Forest on 165,715 High School Students”.

Next Steps: Potential Models & Rationale

- Logistic Regression
  - Estimate the probability that a student belongs to each outcome category.
  - Interpretable baseline showing how variables affect outcome probability.

- Decision Tree
  - Predicts outcomes by splitting students into groups based on independent values.
  - Understandable rules for identifying at-risk students.